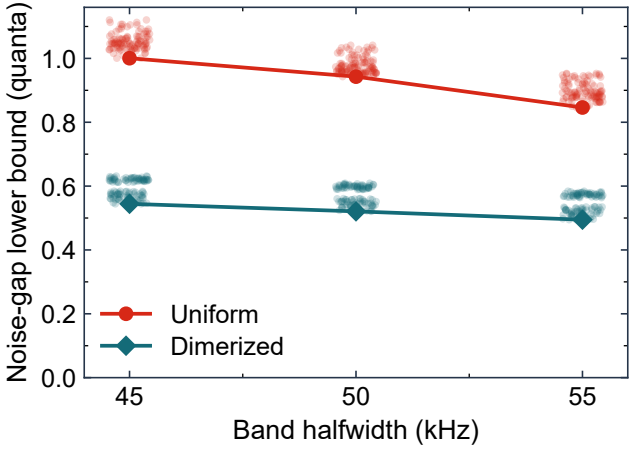
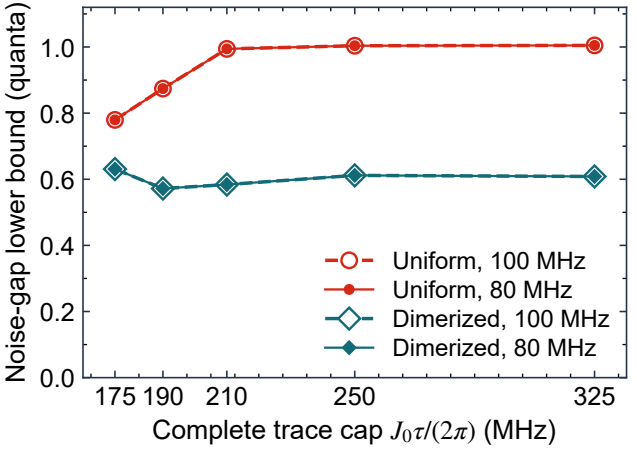
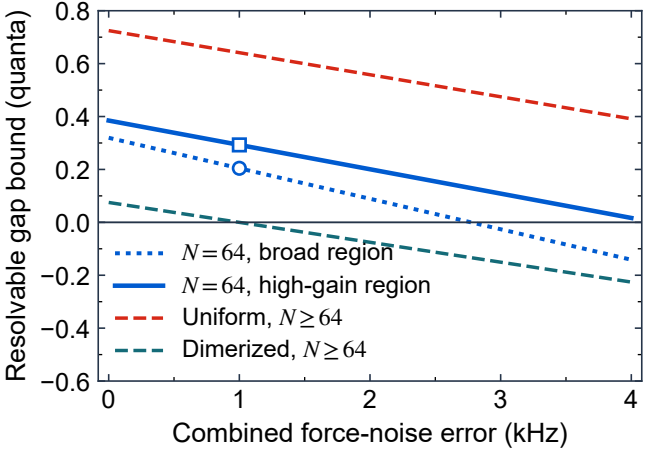

Noise-gap lower bound (quanta)
1.0
0.8
0.6
0.4
0.2
0.0
45
50
55
Band halfwidth (kHz)
Uniform
Dimerized

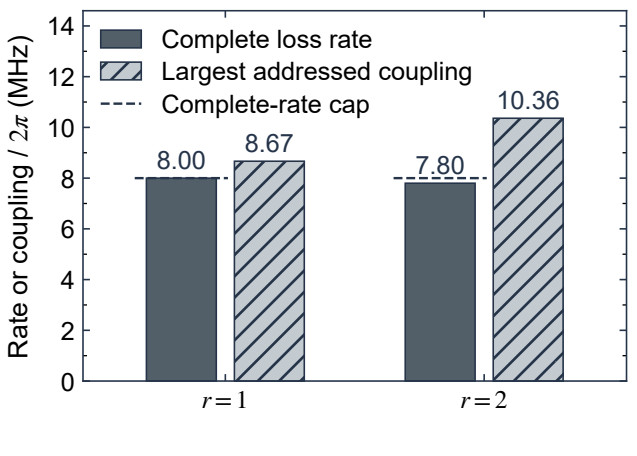
(a)


Noise-gap lower bound (quanta)
1.0
0.8
0.6
0.4
0.2
0.0
175 190
210
250
325
Complete trace cap $J_0\tau/(2\pi)$ (MHz)
Uniform, 100 MHz
Uniform, 80 MHz
Dimerized, 100 MHz
Dimerized, 80 MHz

(b)


Resolvable gap bound (quanta)
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
0
1
2
3
4
Combined force-noise error (kHz)
$N=64$, broad region
$N=64$, high-gain region
Uniform, $N\geq 64$
Dimerized, $N\geq 64$

(c)


Rate or coupling / 2π (MHz)
14
12
10
8
6
4
2
0
Complete loss rate
Largest addressed coupling
Complete-rate cap
8.00
8.67
7.80
10.36
$r=1$
$r=2$

(d)